Locked Out: The Uneven Toll of COVID Border Closures in Sub-Saharan Africa

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Abstract

This investigation examines whether COVID-19 border closures had a disproportionate economic impact on landlocked Sub-Saharan African countries relative to their coastal counterparts. Exploiting the variation within border city pairs—urban centers whose footprints span international borders between one landlocked and one coastal country—a difference-in-differences analysis of nighttime light radiance finds that landlocked sides of border city pairs dimmed significantly following the widespread introduction of border closures in March 2020, relative to their coastal counterparts. This result is robust to controlling for country-specific containment policies and is consistent across alternative fixed effect specifications. Analysis of bilateral export data from UN Comtrade suggests a trade channel: landlocked countries' exports were significantly more sensitive to border closures in neighboring countries than were coastal countries, consistent with their structural dependence on cross-border transit routes for market access. Together, these findings provide some of the first causal evidence that landlocked countries' geographic disadvantage translates into measurably worse economic outcomes when neighboring countries restrict movement.

1 Introduction

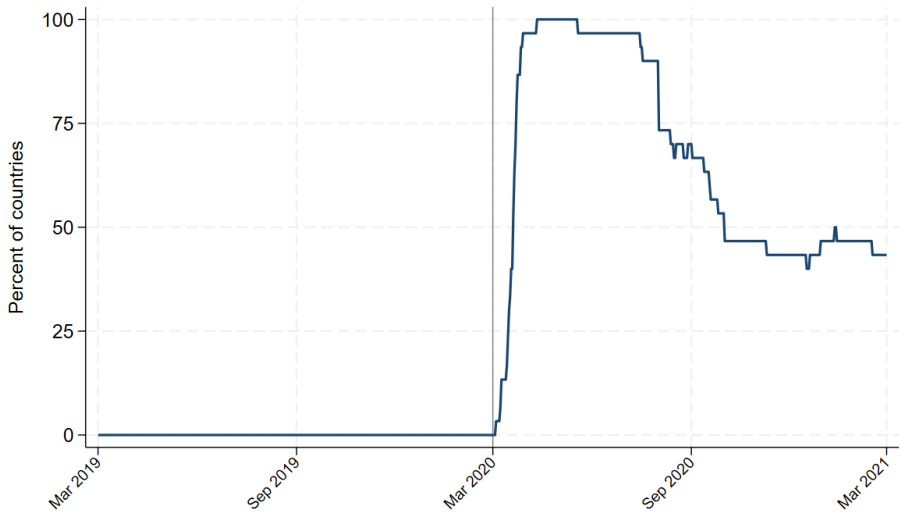


Figure 1: Percent of Sub-Saharan African countries with active COVID border closure policies

Note: Data come from the Covid Border Accountability Accountability Project. Island countries are excluded from the figure as they are not included in later analysis. Closures include both those categorized as "complete" or "partial" closures and exclude country-specific closures directed at non-bordering countries.

Landlocked countries, particularly those in Sub-Saharan Africa are typified by poor access to markets. Even in ordinary times, their participation in international trade, and the development benefits which are associated with trade, rest heavily on factors outside their control: the quality of infrastructure, the stability of governance, and the administrative practices of neighboring coastal countries (Faye et al., 2004). While countries have at times introduced barriers that constrain regional trade, the speed and breadth of restrictions enacted during the early stages of the COVID-19 pandemic were without precedent. Before March 2020, while the spread of COVID-19 had begun in the year prior, no Sub-Saharan African countries had yet enacted border closures[1]; by April 11, every country on the continent had an active border closure, as shown in Figure (1).

As COVID-19 induced waves of border closures across the world, landlocked countries faced disproportionate challenges. Figure (2) gives an estimate of what GDP growth in 2020 was expected in Sub-Saharan African countries prior to the arrival of COVID-19, compared to the most recent estimates. Here, growth forecasts in the January 2020 edition of the Global Economic Prospects report, published by the World Bank, represent an estimate of what market conditions would suggest growth would be in these economies absent the COVID-shock; while COVID-19 had already emerged by the time this report was written, it was months before any Sub-Saharan African countries had closed their borders, and the document does not mention the virus in its

[1]Excluding country-specific bans on travelers from countries outside Sub-Saharan Africa.

text (World Bank, 2020). Compared to such forecasts, Sub-Saharan African countries achieved real GDP growth 6.4 percentage point below what was expected ex-ante given market conditions. Coastal Sub-Saharan African countries, on the other hand, achieved growth 5.2 percentage points below pre-COVID estimates, on average.

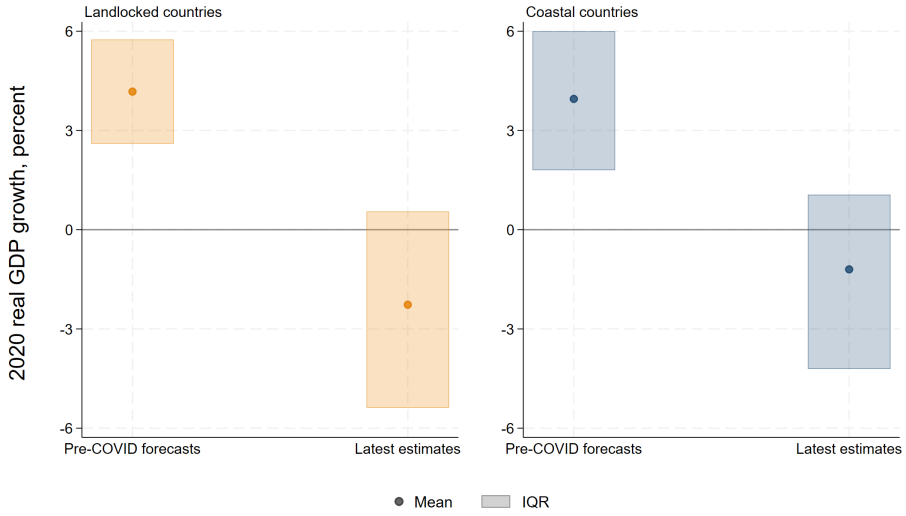


Figure 2: Real GDP growth, January 2020 forecasts vs recent estimates

Note: Bars represent the interquartile range of real GDP growth forecasts/estimates among landlocked and coastal Sub-Saharan African countries, using the World Bank's classification of the Sub-Saharan Africa region. Pre-COVID forecasts are sourced from the World Bank's January 2020 Global Economics Forecast publication, while latest estimates represent those available on the World Bank's World Development Indicators portal. Island nations are excluded from the figure, as they are not included in later analysis. Ethiopia and Uganda are also excluded as World Bank forecasts and estimates are made on a fiscal year basis, which was nearly complete by the arrival of COVID-19 to the continent.

This investigation seeks to better understand whether COVID-19 border closures had a disproportionate economic impact on landlocked countries, relative to their coastal neighbors, and examines the mechanisms by which this difference emerged. While the disproportionate undershooting of forecasts presented in Figure (2) is suggestive of such a disproportionate impact, a more granular measure of economic well-being is necessary for identifying a more direct link between COVID-19 and its disproportionate impact on landlocked countries. To this end, this investigation exploits border city pairs—urban centers whose footprints span international borders between one landlocked and one coastal country—as a natural experiment that isolates the effect of landlocked status from local geographic conditions.

While prior investigations have linked a country's landlocked status to trade dependence on neighboring countries (Limão and Venables, 2001; Faye et al., 2004), the causal content of this relationship is difficult to establish. Landlocked countries differ from coastal ones along many dimensions beyond port access, and the policy barriers they face are typically slow-moving and endogenous to bilateral relationships. The COVID-19 pandemic offers a rare opportunity to study this dependence under quasi-experimental conditions: border closures were rapid, near-universal,

and driven by epidemiological considerations largely external to any bilateral trade relationship. By exploiting this shock alongside the within-urban center variation afforded by these border city pairs, this investigation provides some of the first causal evidence that landlocked countries' structural dependence on neighbors translates into measurably worse economic outcomes when those neighbors restrict movement.

In order to quantify local conditions at such a granular level, remotely sensed nightlights data serves as a proxy for local economic activity. This practice is well-established in the economics literature as a reliable measure of subnational economic output, particularly in data-scarce environments (Henderson et al., 2012). By conducting a difference-in-differences analysis of nightlight brightness, this investigation finds that, the year following the widespread introduction of border closures in March 2020, nightlight brightness on the landlocked side of border cities dimmed relative to their coastal counterparts. This relative dimming is significant even when controlling for city/month-specific border closures, suggesting that even after border closures were lifted in some countries following the initial wave, landlocked countries continued to suffer worse economic conditions.

One possible explanation for the relative decline in nightlight brightness in landlocked countries is the disproportionate impact that local border closures had on landlocked countries' ability to trade with third parties. In other words, for landlocked countries, border closures inhibited not only bilateral trade between the two bordering countries, but also trade with many other markets. Analysis of such a mechanism, using rich bilateral trade data from UN Comtrade, suggests that landlocked countries saw much greater declines in exports in response to neighboring countries erecting border closures than coastal Sub-Saharan African countries did. Together these results suggest that landlocked countries were disproportionately isolated from international markets during the COVID-19 pandemic.

This investigation will proceed as follows: Section 2 outlines the data used in this investigation, Section 3 explains the investigation's methodology, Section 4 outlines results, and Section 5 concludes.

2 Data

To investigate the effects of COVID-19 and its border control measures on local economic conditions, this investigation uses NASA's Black Marble dataset for monthly spatial brightness data (NASA Earth Observatory Group, 2026), and NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) land use product to identify border city pairs along international borders. The COVID Border Accountability Project's border closure database (COVID Border Accountability Project, 2021) is used throughout and additional investigation into the mechanism of international

trade is conducted using data from UN Comtrade (United Nations Statistics Division, 2026). Further robustness checks use policy stringency data from the Oxford COVID-19 Government Response Tracker (Hale et al., 2021).

A. NASA Black Marble

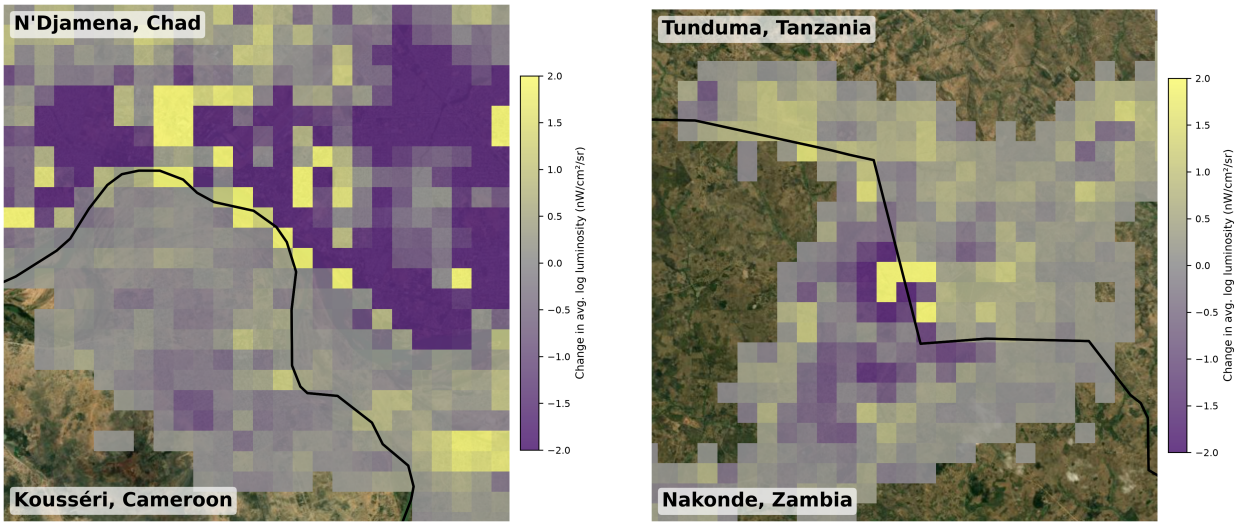


Figure 3: Change in nightlight brightness in select border city pairs

Note: The following represents the difference in average nightlight brightness for the period between March 2020 and February 2021 and the period between March 2019 and February 2020. Pixels with brightness values equal to zero for the duration of the investigation are left fully transparent. Satellite basemap imagery are sourced from Esri World Imagery.

Nighttime light intensity, visible from satellite imagery, provides a uniquely granular and consistent measure of local economic activity across the cities and borders at the heart of this investigation (Henderson et al., 2012). To measure such nighttime light intensity, this investigation uses NASA's Black Marble VNP46A3 product, a monthly composite derived from the Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band aboard the Suomi NPP satellite (NASA Earth Observatory Group, 2026). Black Marble applies a suite of corrections to raw VIIRS observations, to produce calibrated, physically interpretable radiance values. Monthly composites are used rather than daily observations to reduce noise from cloud cover and other transient atmospheric effects. Black Marble data is available starting in 2012, although for all regression analysis, only the period between March 2019 and February 2021 is used.

As off-angle observations may suffer from distortions, and as this investigation relies on a precise understanding of which side of an international border a given nightlight signal originates, the *NearNadir_Composite_Snow_Free* field was chosen, which restricts the composite to observations taken near the satellite's nadir viewing angle. The spatial resolution of the data is 500m x 500m, and average monthly radiance values are measured in units of nW/cm^2/sr.

Urban centers are identified using NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) land use product. The MODIS MCD12Q1 product is an annual 500m x 500m land cover classification derived from the Terra and Aqua satellites, which assigns each pixel to one of seventeen land cover types under the IGBP classification scheme (Friedl and Sulla-Menashe, 2022). Pixels which were identified by MODIS as urban in 2019 which either intersect with an international border between a landlocked and a coastal Sub-Saharan African country, or are directly adjacent to an international border pixel, were identified as a border city for analysis. A 5km radius is drawn around each urban border pixel to define each urban center; in the case where multiple urban border pixels are identified, the union of the 5km radius circles is taken. To ensure that the identified areas on either side of the border exhibit meaningful economic activity, and to exclude edge cases with strange border geometry, this investigation restricts the sample to cities where, in the central core within 2.5km of the urban center's centroid, each side of the border has at least fifty pixels which are flagged as high quality and at least five percent of pixel-months had a luminosity value above a threshold of 1 $nW/cm^2/sr$ in 2019. 23 cities meet these criteria, a list of these cities and their characteristics is provided in Table (A1) in the appendix.

Regression analysis throughout uses log(1 + radiance) as the relevant luminosity outcome variable, and restricts the sample to data flagged as high quality (i.e. without weather-related concerns to data quality). To give equal weight to each country-city, regardless of the number of pixels it contains, regressions weight each pixel by the inverse total number of pixels on its country-city.

Figure (3) demonstrates, for two example border city pairs, how differences in brightness between pre-COVID and COVID eras were disproportionately negative in landlocked countries. Here, the purple pixels, representing declines in nightlight radiance, are concentrated in Chad and Zambia, two landlocked countries, while Cameroon and Tanzania, two coastal African countries, experienced little change.

Figure (4) illustrates normalized nighttime radiance values for both pre-COVID and during the pandemic. While some variance pre-COVID exists due to seasonal trends and as weather events lead to sample distribution changes over months, the landlocked and coastal distributions mirror each other closely. Early in the onset of the COVID pandemic however, the two series diverge and landlocked cities consistently fall below their coastal counterparts, with eight consecutive months in which average coastal radiance exceeded their landlocked counterparts. In April 2020 specifically, the height of prevalence of border closure policies, 79.9 percent of landlocked pixels fell below pre-COVID average levels of log radiance (recorded z-scores below zero), while only 70.5 percent of coastal pixels did the same.

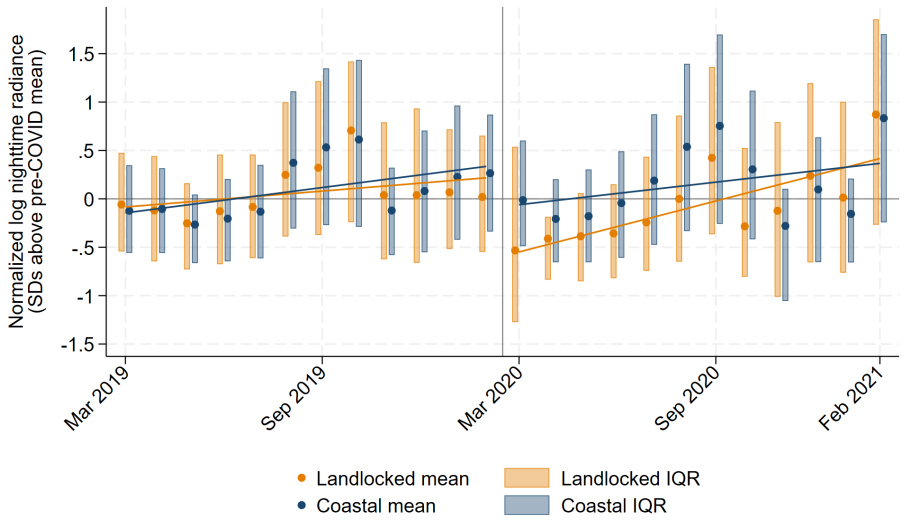


Figure 4: Average normalized nightlight luminosity

Note: Bars represent the interquartile range of pixel-level normalized log nighttime radiance among landlocked and coastal sides of border cities, with each country-city weighted equally. Each pixel's log radiance is normalized by subtracting its mean and dividing by its standard deviation over the March 2017-February 2020 baseline period. Log radiance is computed as log(ntl+1), outlier pixels with normalized z-scores exceeding 10 standard deviations in absolute value are excluded. Only pixels within 5km of the border urban zone and passing quality filters are included. The vertical line marks March 2020. Trend lines represent OLS fits to the pre- and post-March 2020 periods separately.

B. Border Closures

The COVID Border Accountability Project publishes a country-level dataset of travel and immigration bans introduced in response to COVID-19 (COVID Border Accountability Project, 2021). It was led by Dr. Mary Amme-Shiraef and uses policy announcements to construct country-specific timelines of border closure policies for nearly every country globally, including all Sub-Saharan African countries within the purview of this investigation. For the purposes of this investigation, all border closures, besides those which target countries from outside of Sub-Saharan Africa (e.g. China-specific border closures which were common in early 2020) are aggregated to create a country-month-level border closure dummy variable. When aggregating, any month with a border closure active for more than one day is considered as having an active border closure in treatment dummies. This allows for policies which end on the first of the month to be excluded from analysis, while capturing even short closures.

The other dataset of border closures used is the Oxford COVID-19 Government Response Tracker (Hale et al., 2021). This dataset similarly aggregates border closures policies into a convenient dataset. Unfortunately it lacks the complete coverage that the COVID Border Accountability Project has; however, it includes a country-specific stringency index, which quantifies the severity of containment and closure policies more broadly. This index summarizes measures such as school closures, restrictions on public gathering, and closure of public transit, among other components

into one value. This indicator is used in a series of robustness checks, to ensure that the disproportionate dimming in landlocked countries is not an effect of more strict containment measures, such as curfews or stay-at-home measures, which may suppress nighttime lights.

C. UN Comtrade

To investigate the trade channel through which border closures may have affected landlocked countries, this investigation draws on bilateral trade data from the United Nations Comtrade database (United Nations Statistics Division, 2026). Comtrade collects and standardizes annual merchandise trade flows reported by national statistical authorities, covering exports and imports by country pair and product. This investigation uses annual export flows at the 4-digit Harmonized System (HS) level for Sub-Saharan African countries in 2019 and 2020, comparing pre- and post-closure trade patterns across landlocked and coastal reporters. While annual data for 2020 includes some months before most border closures were implemented, it allows for maximum coverage across countries, and serves as a reasonable approximation of the pandemic era[2].

Analysis of Comtrade data uses the inverse hyperbolic sine (IHS) of the primary value of bilateral exports as its outcome variable[3]. IHS is useful, as it approximates a log transformation for large values, but is defined at zero, so if the bilateral trade in a given product category is zero for either year of analysis, it will nonetheless be included in analysis. Regressions weight observations to give equal weight to each reporter country, with partner countries weighted by their share of the reporter's 2019 exports.

3 Methodology

A. Nighttime Lights

To estimate the differential impact of COVID-19 border closures on landlocked countries, this investigation employs a difference-in-differences design exploiting the variation within a given urban area, afforded by cross-border city pairs. The core identifying assumption is that, absent the COVID-19 shock, nighttime light intensity on the landlocked and coastal sides of each border city pair would have followed parallel trends. The border city design lends credibility to this assumption:

[2] 32 continental Sub-Saharan African countries have sufficient trade data in UN Comtrade, 14 of which are landlocked. They are as follows: Angola, Benin, Botswana, Burkina Faso, Burundi, Cameroon, Central African Republic, Côte d'Ivoire, DR Congo, Eswatini, Ethiopia, Gabon, Gambia, Ghana, Kenya, Lesotho, Liberia, Malawi, Mali, Mauritania, Mozambique, Namibia, Niger, Nigeria, Republic of Congo, Rwanda, Senegal, South Africa, Tanzania, Togo, Uganda, Zambia, and Zimbabwe. Ghana, Mozambique, and Zimbabwe have only monthly data available during this period; for those countries, annual data is aggregated from monthly data.

[3] $\text{IHS}(x) = \ln\left(x + \sqrt{x^2 + 1}\right)$

because both sides of a border city pair share the same local geography, and are oftentimes deeply economically integrated, any divergence in luminosity following the onset of border closures can be attributed to differences in how the two countries responded to the pandemic, rather than to pre-existing differences in economic conditions.

$$\log(1 + \text{NTL}_{it}) = \alpha + \beta_1(\text{COVID}_t \times \text{Landlocked}_i) + \beta_2\text{COVID}_t + \rho_i + \varepsilon_{it} \quad (1)$$

In Equation (1), NTL_{it} refers to the nighttime lights radiance in pixel i in month t. COVID_t is a dummy variable equal to one for March 2020 onward, Landlocked_i equals one if pixel i is in a landlocked country, and β_1 is the difference-in-differences regression coefficient of interest. ρ_i represents pixel-level fixed effects, and ε_{it} is the idiosyncratic error term. Standard errors are clustered by border city pair. Note that the Landlocked_i term is omitted from Equation (1) as it would otherwise be absorbed by pixel fixed-effects.

While including a sharp distinction between pre-COVID and COVID eras at March 2020, as indicated by the COVID_t term, is imperfect, the roll out of border closures in March 2020 was rapid, as shown in Figure (1), such that, while no Sub-Saharan African countries had an active border closure policy targeting other African countries on March 1, 2020, all but one (Tanzania) had introduced such a border closure policy by March 31, and by April 11, all Sub-Saharan African countries had a border closure policy in place.

The identifying assumption is that, absent COVID-19 border closures, landlocked and coastal countries would have followed similar trends in nighttime light intensity. The border city pair design supports this assumption: coastal countries can access international markets through their own ports, insulating them from the border closure policies of neighboring countries, while landlocked countries depend on transit through those neighbors. Any divergence in nightlight radiance following March 2020 can therefore be attributed to differential exposure to border closures rather than pre-existing differences in nightlight trajectories.

$$\log(1 + \text{NTL}_{it}) = \alpha + \beta_1(\text{COVID}_t \times \text{Landlocked}_i) + \beta_2\text{COVID}_t + \gamma_c + \delta_p + \varepsilon_{it} \quad (2)$$

Equation (2) mirrors Equation (1) closely, except pixel fixed effects are replaced with country (γ_c) and border city pair (δ_p) fixed effects. It is an alternative specification which is presented alongside results with pixel-level fixed effects.

In both cases, regressions are calculated with analytical weights equal to the inverse number of pixels in each country-city, such that each border city pair receives equal weight, and each side of the border likewise is weighted equally, regardless of the number of observations on either side.

One possible limitation of using nightlights data during the COVID-19 pandemic is that some policies to curb the spread of COVID-19, including curfews or stay-at-home measures, may affect

nightlights independent of its usefulness as a proxy for economic activity. For that reason, regressions using Equation (1) are run just for samples where, of the two countries in the border city pair, the landlocked country's stringency index is higher, and likewise for where the coastal country's stringency index is higher. The assumption underlying this exercise is that, if the relative dimming of landlocked cities is driven by stricter containment measures rather than border closure-induced trade disruption, one would expect the effect of nightlight brightness to be concentrated in city pairs and in months where the landlocked country has the higher stringency index. Finding instead that the effect persists in instances where the coastal country has the higher stringency index would suggest that the result reflects the landlocked country's isolation from international markets, rather than its own domestic containment policies suppressing nighttime activity.

$$\begin{aligned}\log(1 + \text{NTL}_{it}) = \alpha + \beta_1(\text{COVID}_t \times \text{Landlocked}_i) + \beta_2\text{COVID}_t + \beta_3(\text{Closure}_{pt}^{LL} \times \text{Landlocked}_i) \\ + \beta_4\text{Closure}_{pt}^{LL} + \beta_5(\text{Closure}_{pt}^{C} \times \text{Landlocked}_i) + \beta_6\text{Closure}_{pt}^{C} + \rho_i + \varepsilon_{it}\end{aligned} \tag{3}$$

Equation (3) presents the full regression specification, adding border closure terms to Equation (1). Here, Closure_{pt}^{LL} equals one if the landlocked country of border city pair p has an active border closure in month t, and Closure_{pt}^{C} equals one if the coastal country has an active border closure in month t. Each of these variables is a border city pair-level indicator, meaning if the landlocked side of a given border city pair has an active border closure in month t, Closure_{pt}^{LL} will equal one in all pixels in that pair. Closure is equal to zero for all border city pairs for all months prior to March 2020. Closure terms are added sequentially in analysis, such that the first regression is calculated with Closure_{pt}^{LL} terms alone, the second is calculated with Closure_{pt}^{C} terms alone, and the final regression is calculated using the full specification of Equation (3).

B. Export Channel

To examine the trade channel through which border closures may have suppressed economic activity in landlocked countries, this investigation uses bilateral export data from UN Comtrade to estimate how exports from landlocked and coastal Sub-Saharan African countries responded to COVID-19 border closures.

$$\text{IHS}(\text{X}_{ijkt}) = \alpha + \beta_1(\text{COVID}_t \times \text{Landlocked}_i) + \beta_2\text{COVID}_t + \gamma_{ij} + \delta_k + \varepsilon_{ijkt} \tag{4}$$

Regressions using Equation (4) represent the initial analysis on the effects of COVID-19 on exports from landlocked countries. Here, $\text{IHS}(\text{X}_{ijkt})$ is equal to the inverse hyperbolic sine of

the primary value of exports from origin country i to destination country j for product k in year t. Unlike for equations using spatial nightlight radiance, where January and February 2020 were considered pre-treatment, COVID_t in this instance equals one for the year 2020, and zero for the year 2019, as trade data are annual. γ_{ij} are fixed effects for origin-destination country pairs, δ_k are fixed effects for the product category using 4-digit Harmonized System (HS) codes. β_1 is the difference-in-differences estimate of the effects of COVID on landlocked Sub-Saharan African countries. Standard errors are clustered at the origin country-level, and analytical weights are constructed such that each origin is weighted equally, and, for a given origin country, recipient countries are weighted as a proportion of 2019 trade with the given recipient.

$$\begin{aligned}\text{IHS}(\text{X}_{ijkt}) =& \alpha + \beta_1(\text{COVID}_t \times \text{Landlocked}_i) + \beta_2 \text{COVID}_t \\ &+ \beta_3 \text{C}_{it}^{border} + \beta_4(\text{C}_{it}^{border} \times \text{Landlocked}_i) \\ &+ \beta_5 \text{C}_{it}^{origin} + \beta_6(\text{C}_{it}^{origin} \times \text{Landlocked}_i) \\ &+ \beta_7 \text{C}_{jt}^{recip} + \beta_8(\text{C}_{jt}^{recip} \times \text{Landlocked}_i) + \gamma_{ij} + \delta_k + \varepsilon_{ijkt}\end{aligned} \tag{5}$$

Equation (5) presents the full specification in the results table in Columns (2)-(5) in the exports subsection of this investigation's Analysis section, but with border closure terms introduced sequentially. C_{it}^{origin} and C_{jt}^{recip} represent the number of days in year t in which origin country i and recipient country j respectively have active border closures. C_{it}^{border} is analogous to similar indicators, but for bordering countries. For each country bordering country i, the number of days in which that country has an active border closure is determined; that value is averaged across all bordering countries, with countries weighted proportional to the total value of exports from origin country i in 2019, as a proportion of its total exports to bordering countries. In other terms, C_{it}^{border} is the trade-weighted average number of days in 2020 that bordering countries had active border closures. All such closure variables were constructed using data from the COVID Border Accountability Project. By construction, as no border closures were in place in 2019, all closure variables equal zero for 2019.

The assumption underlying the difference-in-differences construction is that, for most international trade, coastal countries are uninhibited by border closures in neighboring countries, as ports along their own coasts allow for accessing international markets independent of neighboring countries' policies, and that, absent border closures, landlocked and coastal countries would face similar trends in international exports. β_1 in this equation represents the principle difference-in-differences estimator of this equation. Additionally, of particular interest is the coefficient β_4, which captures the differential sensitivity of landlocked countries to border closures in neighboring countries relative to coastal countries. A negative coefficient estimate for this term would suggest that

landlocked countries experience a greater reduction in exports in response to bordering countries' border closures than coastal countries.

4 RESULTS

The results presented in this section suggest that COVID-19 border closures had a measurably larger negative impact on economic activity in landlocked countries than in their coastal counterparts. Nighttime light radiance on the landlocked side of border city pairs dimmed significantly relative to the coastal side following March 2020, and this differential is robust to controlling for the specific border closure policies of each country. Comtrade data corroborate this finding at the macroeconomic level, showing that landlocked countries experienced substantially larger declines in exports in response to neighboring countries' border closures.

A. Main Results

Table 1: Difference-in-Differences: Log nightlight radiance

	All Cities		**Excluding DRC Border Cities**	
	(1)	(2)	(3)	(4)
COVID	0.0284**	0.0350*	0.00463	0.0240
	(0.0125)	(0.0170)	(0.0108)	(0.0170)
COVID × *Landlocked*	−0.0421***	−0.0559***	−0.0253*	−0.0437**
	(0.0124)	(0.0198)	(0.0132)	(0.0171)
Constant	0.346***	0.0556***	0.421***	0.0550***
	(0.00447)	(0.00733)	(0.00513)	(0.00951)
Observations	123,706	124,275	84,524	84,696
R-squared	0.950	0.244	0.964	0.283
Fixed effects	Pixel	Country + City Pair	Pixel	Country + City Pair

Note: A small number of pixels have high-quality observations in only a single month. These singleton pixels are automatically dropped by the estimation routine when pixel fixed effects are included. This accounts for the small discrepancies in observation counts between analogous regressions.
Robust standard errors in parentheses.
*** p<0.01, ** p<0.05, * p<0.1

Table (1) shows results from Equations (1) and (2), which are this investigation's principal regression results. Regression Column (1) demonstrates that landlocked cities experienced a decrease in log radiance by 0.0421, which is significant at the 1 percent level. Findings are similarly strong with country and border city pair fixed effects. Counterintuitively, the coastal

portion of border city pairs became increasingly bright after March 2020, as the positive COVID terms suggest.

It could be argued that including city pairs along the border of the Democratic Republic of the Congo distort results. The main mechanism argued throughout this investigation, that COVID border closures disproportionately affected landlocked cities' access to markets, is arguably weak in cities bordering the DRC, given the physical distance from its eastern borders and its limited coastline. For this reason, regression estimates excluding such border city pairs are similarly presented. Encouragingly, results with negative difference-in-differences coefficients for landlocked countries hold in all regression specifications, albeit with weakened statistical significance. These results also partly explain the unexpected statistically significant COVID coefficient, as coefficient terms for COVID become statistically insignificant when limiting the sample to exclude the DRC.

One alternative explanation for the disproportionate decline in nighttime luminosity in landlocked countries is that landlocked countries implemented stricter domestic COVID containment measures, such as curfews or stay-at-home orders, which could suppress nighttime lights independent of any decline in economic activity. To control for this, the Oxford COVID-19 Government Response Tracker's stringency index is used. This indicator summarizes a suite of containment measures, including school closures, restrictions on public gathering, and closure of public transit, in order to quantify the strictness by which COVID containment policies are implemented. In Table (2), regression (1) mirrors the principal regression from Table (1), for ease of comparison. Regression (2) is run in a sample including all border city pair-months prior to March 2020, and all subsequent months in which the landlocked country had a higher stringency index. Regression (3) is similar, but includes only border city pair-months in which the coastal country had a higher stringency index. Consistent with a finding that border control measures disproportionately negatively affect landlocked countries, result estimates remain negative and statistically significant regardless of which country has the higher stringency index.

Columns (4)-(6) control for border closures with border city pair-month-specific dummy variables. Results show that, regardless of the border control measures taken, landlocked countries saw disproportionately negative changes in nighttime light radiance. Interestingly, there is suggestive evidence that border closure policies negatively affect the side of the border which did not implement the policy. A statistically significant negative coefficient for the *Coastal Closure x Landlocked* and positive term for *LL Closure x Landlocked* in Regression (6) indicates that landlocked countries were more negatively impacted by coastal countries' border closures, and less impacted by their own closures.

Table 2: Difference-in-Differences: Stringency Controls and Border Closure Measures

	Stringency Subsamples			**Border Closure Controls**		
	(1) *Baseline*	(2) *LL stringent*	(3) *Coastal stringent*	(4) *Coastal closure*	(5) *LL closure*	(6) *Both*
COVID	0.0284**	0.0515**	0.0133	0.0449**	0.0744***	0.0664**
	(0.0125)	(0.0189)	(0.0114)	(0.0208)	(0.0237)	(0.0248)
COVID × *Landlocked*	−0.0421***	−0.0507***	−0.0322**	−0.0372**	−0.0625***	−0.0503***
	(0.0124)	(0.0165)	(0.0128)	(0.0169)	(0.0167)	(0.0172)
Coastal Closure				−0.0223		0.0193
				(0.0220)		(0.0152)
Coastal Closure × *Landlocked*				−0.00933		−0.0333*
				(0.0163)		(0.0167)
LL Closure					−0.0661**	−0.0751***
					(0.0261)	(0.0241)
LL Closure × *Landlocked*					0.0286	0.0453**
					(0.0182)	(0.0190)
Constant	0.346***	0.365***	0.345***	0.347***	0.346***	0.346***
	(0.00447)	(0.00383)	(0.00376)	(0.00435)	(0.00409)	(0.00408)
Observations	123,706	80,316	99,807	123,706	123,706	123,706
R-squared	0.950	0.953	0.952	0.950	0.951	0.951
Fixed effects	Pixel	Pixel	Pixel	Pixel	Pixel	Pixel

Robust standard errors in parentheses.
*** p<0.01, ** p<0.05, * p<0.1

B. Export Channel Mechanism

One clear mechanism by which nightlight results might have emerged is through a decrease in international trade. As border closures came into effect across Sub-Saharan Africa, a coastal country with direct access to ports is generally unimpeded in their access to markets, outside of border closures in recipient countries and any restrictions imposed domestically. Landlocked countries, however, are beholden to their neighbors' policies in order to access ports and export goods.

Table 3: Border Closures and Exports

	(1)	(2)	(3)	(4)	(5)
Landlocked × *COVID*	−0.0657	6.297**	5.737***	6.560**	5.901***
	(0.125)	(2.464)	(1.200)	(2.616)	(1.413)
Landlocked		−0.0255**	−0.0170***	−0.0247**	−0.0165***
× *Neighbor closed*		(0.00974)	(0.00423)	(0.00958)	(0.00447)
Landlocked			−0.00585*		−0.00581*
× *Origin closed*			(0.00325)		(0.00323)
Landlocked				−0.00172	−0.00119
× *Recipient closed*				(0.00303)	(0.00242)
Constant	9.755***	9.756***	9.757***	9.756***	9.757***
	(0.0306)	(0.0244)	(0.0165)	(0.0245)	(0.0168)
Observations	255,264	255,264	255,264	255,160	255,160
R-squared	0.363	0.363	0.363	0.363	0.363
Fixed Effects	Prod. + Pair	Prod. + Pair	Prod. + Pair	Prod. + Pair	Prod. + Pair

Robust standard errors in parentheses.
*** p<0.01, ** p<0.05, * p<0.1

Table (3) shows results from Equations (4) and (5). Here, *COVID* is a dummy variable equal to one in the year 2020, and zero in 2019, and the three *Closed* variables represent the number of days in 2020 in which the given country had an active border closure policy. While baseline results in Column (1) show no difference in landlocked countries, exports during 2020, the coefficient for the interaction term *Landlocked x Neighbor closed* is negative and statistically significant in all four equations in which it is included, which is suggestive of a strong dependence of exports in landlocked countries on bordering countries' border closure policies. Specifically, the coefficient term in Equation (5) suggests that, for each additional day in which neighboring countries have active border closures in place, landlocked countries experience a 0.0165 reduction in IHS exports, which approximates a 1.5 percent reduction in exports by value. This effect is quite large and points to the degree to which landlocked countries' exports are subject to the policy decisions of their neighboring countries.

5 Conclusion

This investigation provides causal evidence that COVID-19 border closures had a disproportionately negative impact on economic activity in landlocked Sub-Saharan African countries relative to their coastal counterparts. Exploiting the variation within border city pairs (urban centers whose footprints span international borders between one landlocked and one coastal country), this investigation finds that, relative to their coastal counterparts, nighttime light radiance on the landlocked side of border city pairs declined significantly following the widespread introduction of border closures in March 2020. This relative dimming is robust to controlling for country-specific containment policies, suggesting that the effect reflects landlocked countries' structural dependence on neighboring countries' market access rather than their own domestic policy responses.

Analysis of bilateral export data from UN Comtrade corroborates this finding at the macroeconomic level. Landlocked countries' exports were significantly more sensitive to border closures in neighboring countries than those of coastal countries, consistent with the interpretation that border closures disrupted the transit routes on which landlocked countries depend for access to international markets. Together, these results suggest that the disproportionate economic underperformance of landlocked Sub-Saharan African countries during the COVID-19 pandemic was not merely coincidental, but reflects a specific and measurable channel: the severing of cross-border transit links on which landlocked economies structurally depend.

These findings carry implications for both policy and future research. From a policy perspective, they underscore the importance of maintaining transit corridors for landlocked countries during global crises, and suggest that blanket border closure policies impose asymmetric costs that fall disproportionately on the most trade-dependent economies. From a research perspective, this investigation demonstrates that border city pairs offer a promising quasi-experimental design for isolating the effects of policy shocks that vary by country, and that nighttime lights data can detect economically meaningful divergence at a granular spatial scale even over relatively short time horizons.

This investigation contributes to a growing body of evidence that landlocked countries' vulnerability to their neighbors' policies is not merely a structural background condition, but an active source of economic risk that can be triggered by external shocks. As the global community considers how to design more resilient trade systems in the aftermath of COVID-19, the experience of landlocked Sub-Saharan Africa offers a stark reminder of how quickly geographic disadvantage can become acute economic hardship.

6 Appendix

Table A1: Border City Pair Sample Characteristics

Landlocked	Coastal	Lat	Lon	*N* LL	*N* Coastal
TCD	CMR	12.097	15.018	565	333
BFA	GHA	11.001	-2.757	153	221
ETH	SOM	9.603	43.327	181	229
CAF	COD	4.365	18.636	152	117
ETH	KEN	3.530	39.054	270	271
UGA	KEN	0.464	34.098	252	154
RWA	COD	-1.679	29.247	332	335
RWA	COD	-2.522	28.876	270	470
ZMB	TZA	-9.306	32.763	288	250
ZMB	COD	-10.485	28.645	426	396
ZMB	COD	-11.770	27.238	312	249
ZMB	COD	-12.258	27.996	247	261
ZMB	COD	-12.314	27.959	152	267
ZMB	COD	-12.433	28.336	173	284
ZMB	COD	-12.748	28.475	157	143
ZMB	NAM	-17.489	24.280	141	239
ZWE	ZAF	-22.223	29.984	264	241
BWA	ZAF	-24.881	25.876	218	186
BWA	ZAF	-25.450	23.706	206	272
BWA	ZAF	-25.820	24.678	132	298
SWZ	ZAF	-27.320	31.886	218	242
LSO	ZAF	-28.888	27.894	275	327
LSO	ZAF	-29.292	27.490	423	496

Note: Each row represents one border city pair. Landlocked and Coastal columns report the ISO 3166-1 alpha-3 country codes for each side of the border. Mean latitude and longitude are averaged across all pixels on both sides of the border city pair. *N* reports the number of unique pixels on each side of the border.

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